

EDUSAMBAM ASSESSMENT SERIES

Biology Comprehensive Practice Test 5

PRINTABLE PRACTICE WORKSHEET

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Subject	Biology	Topic	Biology Comprehensive Practice Test 5	Total Marks	100
Total Questions	45	Time Allowed	1 Hour (60 minutes)	Version	A

GENERAL INSTRUCTIONS

1. Read each question carefully before answering.
2. For Section A and Section B, mark one answer only.
3. For Sections C and E, write your answer clearly in the space provided.
4. For Section D, write the correct letter from Column B against each item in Column A.
5. Do not use correction fluid. Keep all answers clear and legible.
6. This worksheet is prepared from the EDUSAMBAM practice assessment supplied for this paper.

PAPER PATTERN

Section	Questions	Marks Each	Total
Section A — Multiple Choice	15	2	30
Section B — True / False	10	2	20
Section C — Fill in the Blank	10	2	20
Section D — Column Matching	5 pairs	2	10
Section E — Short Answer	5	4	20
Total	45	—	100

Section A — Multiple Choice

30 marks

1. The nucleus of a eukaryotic cell contains:

- A.** Chromosomal DNA
C. Only lipids

- B.** Only ATP
D. Cell wall material

Answer: ☐A ☐B ☐C ☐D

Basic • 2 pts

2. A chromosome is best described as:

- A.** DNA associated with proteins and organised into a structural unit
C. A membrane-bound organelle without DNA

- B.** A type of carbohydrate
D. A single protein

Answer: ☐A ☐B ☐C ☐D

Basic • 2 pts

3. A gene is best understood as:

- A.** A functional DNA sequence or region contributing to a biological product or regulatory function
C. A cell organelle

- B.** An entire organism
D. A lipid droplet

Answer: ☐A ☐B ☐C ☐D

Basic • 2 pts

4. DNA replication is described as semiconservative because:

- A.** Each daughter DNA molecule contains one parental strand and one newly synthesised strand
C. No parental DNA is retained

- B.** Both strands are entirely new
D. Only proteins are copied

Answer: ☐A ☐B ☐C ☐D

Basic • 2 pts

5. Meiosis is important in sexual reproduction because it:

- A.** Produces haploid cells and contributes to genetic variation
C. Stops chromosome segregation

- B.** Produces identical diploid body cells only
D. Eliminates all variation

Answer: ☐A ☐B ☐C ☐D

Intermediate • 2 pts

6. A mutation is: A. A change in genetic material sequence or structure C. Always beneficial Answer: ☐ A ☐ B ☐ C ☐ D	B. Always harmful D. Always caused by exercise	Intermediate • 2 pts
7. During gene expression, DNA information can be used to produce: A. RNA and, for protein-coding genes, proteins C. Only carbohydrates Answer: ☐ A ☐ B ☐ C ☐ D	B. Only lipids D. Only ATP	Intermediate • 2 pts
8. Crossing over during meiosis can: A. Exchange DNA between homologous chromosomes and increase variation C. Stop DNA replication permanently Answer: ☐ A ☐ B ☐ C ☐ D	B. Double chromosome number in every gamete D. Create a new organ	Intermediate • 2 pts
9. If two heterozygous parents have a recessive allele for a trait, a child can inherit: A. Two recessive alleles C. No alleles Answer: ☐ A ☐ B ☐ C ☐ D	B. Only dominant alleles D. Four chromosomes from each parent	Intermediate • 2 pts
10. A phenotype is influenced by: A. Genotype and environment, depending on the trait C. Environment only in every case Answer: ☐ A ☐ B ☐ C ☐ D	B. Genes only in every case D. Chromosome number only	Intermediate • 2 pts
11. A mutation in a non-coding regulatory region can matter because: A. It may alter when, where or how strongly a gene is expressed C. It automatically creates a new species Answer: ☐ A ☐ B ☐ C ☐ D	B. Non-coding DNA can never have function D. It always changes the amino-acid sequence	Intermediate • 2 pts
12. Genetic variation is important to evolution because: A. It provides differences on which evolutionary processes can act C. It makes all individuals identical Answer: ☐ A ☐ B ☐ C ☐ D	B. It prevents inheritance D. It stops natural selection	Advanced • 2 pts
13. A karyotype is useful for examining: A. Chromosome number and large-scale chromosome structure C. Only enzyme activity Answer: ☐ A ☐ B ☐ C ☐ D	B. Only blood glucose D. Only species behaviour	Advanced • 2 pts
14. If a cell fails to separate chromosomes correctly during meiosis, the gametes may: A. Receive abnormal chromosome numbers C. Lose all DNA Answer: ☐ A ☐ B ☐ C ☐ D	B. Become bacteria D. Become identical to the parent	Advanced • 2 pts
15. Which statement best integrates heredity? A. DNA is organised into chromosomes; genes are expressed through regulated processes; meiosis and fertilisation transmit and reshuffle genetic information C. Chromosomes are made only of RNA Answer: ☐ A ☐ B ☐ C ☐ D	B. Genes act independently of cells D. Environment has no role in phenotype	Advanced • 2 pts

Section B — True / False

20 marks

16. DNA contains information encoded in the sequence of its nucleotides. A. True ☐ B. False ☐	Basic • 2 pts
17. Every mutation causes a visible disease. A. True ☐ B. False ☐	Basic • 2 pts
18. Somatic cells and gametes normally have different chromosome-number states in organisms with sexual reproduction. A. True ☐ B. False ☐	Basic • 2 pts
19. Meiosis normally produces genetically identical daughter cells. A. True ☐ B. False ☐	Basic • 2 pts
20. Gene expression can be regulated. A. True ☐ B. False ☐	Intermediate • 2 pts
21. A dominant allele is always more common in a population than a recessive allele. A. True ☐ B. False ☐	Intermediate • 2 pts

22. Crossing over occurs between homologous chromosomes during meiosis.

A. True ☐ B. False ☐
23. Mutations can be neutral, harmful or beneficial depending on context.

A. True ☐ B. False ☐
24. A person's genotype and phenotype are always identical concepts.

A. True ☐ B. False ☐
25. Genetic information can influence traits without being the only influence on every trait.

A. True ☐ B. False ☐

Section C — Fill in the Blank

20 marks

26. The hereditary molecule DNA is organised into structures called ____.

Answer: _____
27. A functional region of DNA is commonly called a ____.

Answer: _____
28. DNA copying before cell division is called DNA ____.

Answer: _____
29. The division process that produces haploid gametes is ____.

Answer: _____
30. The physical appearance or measurable characteristics of an organism form its ____.

Answer: _____
31. The genetic constitution of an organism is its ____.

Answer: _____
32. A change in DNA sequence is a ____.

Answer: _____
33. The process by which information in a gene is used to make a functional product is gene ____.

Answer: _____
34. Exchange of DNA between homologous chromosomes is called ____ over.

Answer: _____
35. A cell containing two sets of homologous chromosomes is described as ____.

Answer: _____

Section D — Column Matching

10 marks

36-40. Match each term to its correct definition.

Column A	Answer	Column B
DNA	_____	A. Functional DNA region
Gene	_____	B. Organised DNA-protein structure
Chromosome	_____	C. Produces haploid cells and reshuffles variation
Meiosis	_____	D. Change in genetic material
Mutation	_____	E. Stores hereditary information

Section E — Short Answer

20 marks

41. Explain the relationship among DNA, genes and chromosomes.

Advanced · 4 pts

42. Why is meiosis important for both chromosome-number stability and genetic variation in sexual reproduction?

Advanced · 4 pts

43. Explain why a mutation is not automatically harmful.

Advanced · 4 pts

44. How can two people with similar genes show different phenotypes?

Advanced · 4 pts

45. A child has a recessive condition while both parents are unaffected. Explain a plausible inheritance pattern.

Advanced · 4 pts

Candidate Signature: _____ **Invigilator Signature:** _____

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Marks Obtained: _____ / 100

Checked By: _____

Result / Grade: _____

Date Checked: _____

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Biology Comprehensive Practice Test 5 — Answer Key

Teacher reference: equivalent grammatically correct wording is acceptable for open-ended items when the meaning is preserved.

Section A — Multiple Choice

- | | |
|------|-------|
| 1. A | 9. A |
| 2. A | 10. A |
| 3. A | 11. A |
| 4. A | 12. A |
| 5. A | 13. A |
| 6. A | 14. A |
| 7. A | 15. A |
| 8. A | |

Section B — True / False

- | | |
|-----------|-----------|
| 16. False | 21. True |
| 17. True | 22. False |
| 18. False | 23. False |
| 19. True | 24. True |
| 20. False | 25. False |

Section C — Fill in the Blank

- | | |
|-----------------|----------------|
| 26. chromosomes | 31. genotype |
| 27. gene | 32. mutation |
| 28. replication | 33. expression |
| 29. meiosis | 34. crossing |
| 30. phenotype | 35. diploid |

Section D — Column Matching

- | | |
|--------------------|------------------|
| 36. DNA — E | 39. Meiosis — C |
| 37. Gene — A | 40. Mutation — D |
| 38. Chromosome — B | |

Section E — Short Answer (Model Answers)

- 41.** chromosomes; organise; proteins; genes; functional
- 42.** reduction; haploid; cells; variation; through
- 43.** location; effect; gene; product; regulation
- 44.** environmental; effects; gene; regulation; developmental
- 45.** carrier; parents; recessive; inheritance; child